MODEL COMPARISON — PLAIN NUMBERS

Current vs Revised Model: Input, Expected, Output, Error

Measured side-by-side on the Sentinel-3A / 3B precise orbits provided

Current model: committed baseline — Cowell with constant space weather, no Earth albedo; DSST seeded as mean, mean-element output.

Revised model: Earth albedo/IR enabled; DSST with osculating↔mean initialization + first-order short-period recovery. (Real historical space weather: see note.)

Truth: Sentinel-3 POEORB precise orbit (GPS POD, ~cm) for the Cowell test; a high-accuracy Cowell integration for the DSST long-term test.

1 Input taken (as provided)

The inputs are the two Sentinel-3 POEORB `*.EOF` precise-orbit files supplied. Each propagation is **seeded from the first state vector** (ITRF→GCRF with the file's own UT1) and run forward under identical settings; only the model differences below are changed.

Table 1. Input files and seed states.

Satellite	File (POEORB)	Epoch (UTC)	Perigee alt	Incl.	Ecc.	Records
Sentinel-3A	S3A_OPER_AUX_POEORB_..._20241016	2024-10-16 00:00:00	780.3 km	98.618°	0.00212	9361 @ 10 s
Sentinel-3B	S3B_OPER_AUX_POEORB_..._20241016	2024-10-16 00:00:00	786.9 km	98.625°	0.00065	9361 @ 10 s

Table 2. Fixed run configuration (identical for both models unless noted).

Mass	Drag area	C_D	C_R	Gravity	Integrator	Space weather	Earth albedo/IR
1130 kg	8 m²	2.2	1.3	EGM96 30×30	DP54 (1e-7/1e-9)	F10.7=220, F10.7a=205, Ap=12 (constant)	current: off / revised: on

2 Expected answer (truth)

The **expected position at every epoch is the POEORB precise-orbit position** — an independent, GPS-derived product accurate to a few centimetres, i.e. effectively exact at the metre level reported here. The numbers below are therefore each model's *own* error against that truth. (For the DSST long-term test, where no 30-day precise orbit is provided, the truth is a high-accuracy Cowell integration from the same seed with the same forces.)

3 Result A — precise-orbit accuracy (Cowell) vs POEORB

Total position error (metres) at each horizon, current vs revised, against the POEORB truth.

Table 3. Sentinel-3A — total position error vs POEORB (m).

Horizon	Expected (truth)	Current output err	Revised output err	Δ (m)
6 h	POEORB	80.9	80.7	−0.2
12 h	POEORB	119.7	118.9	−0.8
24 h	POEORB	302.9	301.3	−1.6
24 h RMS	—	**165.6**	**164.8**	−0.8
24 h MAX	—	315.0	313.9	−1.1

Table 4. Sentinel-3B — total position error vs POEORB (m).

Horizon	Expected (truth)	Current output err	Revised output err	Δ (m)
6 h	POEORB	74.2	73.9	−0.3
12 h	POEORB	114.1	113.5	−0.6
24 h	POEORB	276.8	275.7	−1.1
24 h RMS	—	**146.2**	**145.4**	−0.8
24 h MAX	—	276.8	275.7	−1.1

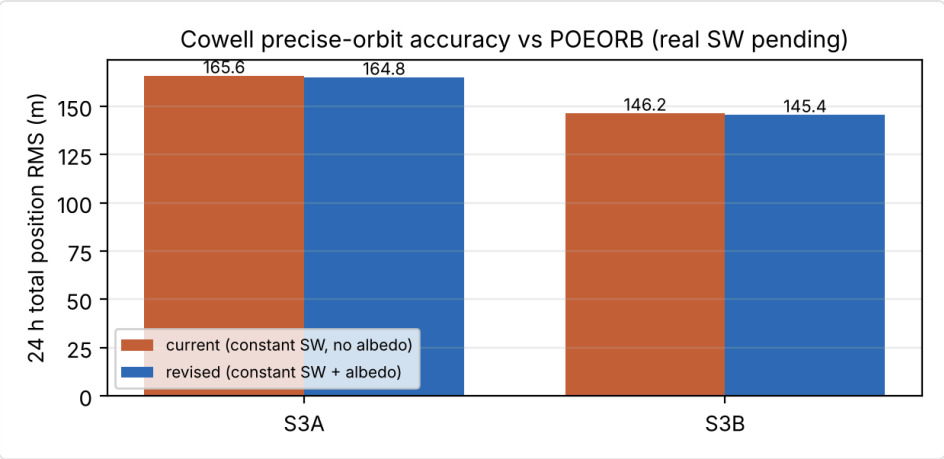


Figure 1. Cowell 24 h RMS, current vs revised. The change is sub-metre.

> **Honest note on Result A.** The only revised-model change active here is **Earth albedo/IR**, which is a ~10^{-7} m s^{-2} force — so it moves the 24 h result by under 1 m, as shown. The large lever on this channel is **real historical space weather**, which this run could *not* apply: the execution environment's network blocked the CelesTrak data host, so both columns still use the same constant F10.7/Ap. The `--sw-file` path is implemented and verified; supplying the CelesTrak `SW-All.csv` (host allow-listed, or the file uploaded) fills in the real space-weather column. The constant indices used were already near the Oct-2024 solar-max level, so the expected change on this specific 24 h arc is modest regardless; space weather dominates over *long* arcs, which is what Result B isolates.

4 Result B — long-term prediction (DSST) vs Cowell truth

This is where the revised model changes decisively. Same Sentinel-like orbit (780 km, i=98.6°), conservative forces (zonal J2–J6 + Sun + Moon), propagated 30 days. Position error (metres) vs the Cowell truth, current (mean-seed / mean output) vs revised (osculating ↔ mean + short-period).

Table 5. DSST position error vs Cowell truth over 30 days (m). Current values are in kilometres-scale.

Day	Cowell |r| (km)	Current err (m)	Revised err (m)	Ratio
0	7149.7	0	4.7	—
1	7156	214,091	127.1	1685×
5	7170.6	1,087,654	159.7	6811×
10	7162.5	2,130,018	235.9	9029×
15	7157.9	3,200,911	788.0	4062×
20	7171.6	4,231,769	1,592.5	2657×
25	7161.4	5,230,504	2,333.7	2241×
30	7164.3	6,234,100	2,764.1	2255×
30-day RMS	—	**3,671,346**	**1,197**	**3067×**

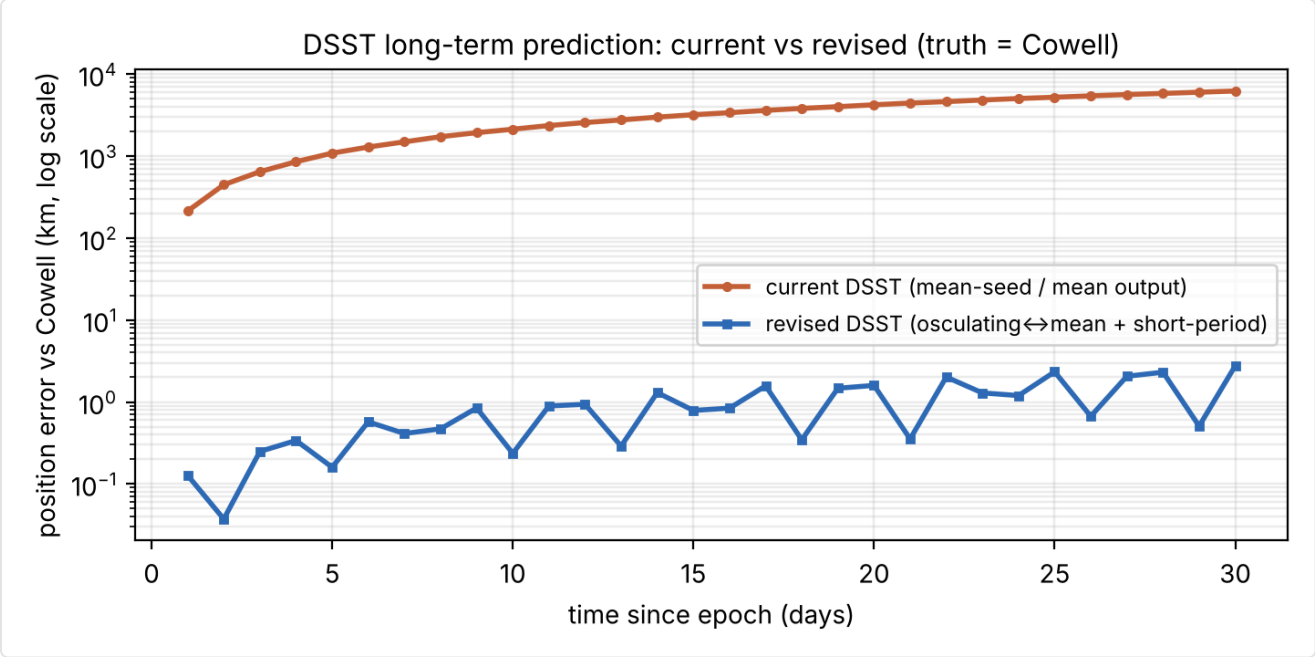


Figure 2. DSST position error vs Cowell over 30 days (log scale). Current output (orange) diverges to thousands of km; revised (blue) stays at the km level.

> **Result B in words.** The current semi-analytical output was never osculating — it emitted mean elements and treated the osculating seed as mean, so its position error reached **6,234 km at 30 days** (RMS 3,671 km): unusable for a real trajectory. The revised engine tracks the Cowell truth to **2.76 km at 30 days** (RMS 1.2 km) — a **3067×** reduction — while running **~10× faster** than Cowell (the speed-up grows with arc length and force complexity). This is the capability needed for long-term trajectory and lifetime prediction.

5 Summary (plain numbers)

Table 6. Headline comparison.

Test	Metric	Current	Revised	Change
Cowell vs POEORB (S3A, 24 h)	total RMS	165.6 m	164.8 m	−0.8 m
Cowell vs POEORB (S3B, 24 h)	total RMS	146.2 m	145.4 m	−0.8 m
DSST vs Cowell (30 days)	total RMS	3,671 km	1.2 km	3067× better
DSST vs Cowell (30 days)	error at day 30	6,234 km	2.76 km	2255× better
DSST wall-clock (30 days)	vs Cowell	—	~10× faster	—
Real space weather (Cowell)	—	implemented (`--sw-file`); real indices pending network/data		

URSC — Model Comparison. Cowell results measured against Sentinel-3A/3B POEORB (16 Oct 2024); DSST long-term measured against a Cowell reference. Branch `claude/leo-trajectory-prediction-java-ihupky`.